Week-5

1# Perform a basic SWOT analysis (Strengths, Weaknesses, Opportunities, Threats) for a sample organization's IT security setup.

Basic SWOT Analysis of a Sample Organization's IT Security Setup

Sample Organization Context: A mid-sized company with:

- Firewall protection
- Antivirus software on endpoints
- Basic password policies
- Cloud-based email system
- Small internal IT team

Strengths

- **Firewall Protection:** Prevents unauthorized external access.
- **Antivirus/Endpoint Protection:** Protects against common malware and viruses.
- **Regular Data Backups:** Reduces risk of permanent data loss.
- **Cloud Email Security:** Cloud provider offers spam filtering and encryption.
- **Basic Access Controls:** Password authentication for internal systems.

Weaknesses

- **Weak Password Policies:** No enforcement of multi-factor authentication (MFA).
- **Limited Security Monitoring:** No dedicated Security Operations Center (SOC).
- **Outdated Software:** Some systems not regularly patched.
- **Lack of Security Awareness Training:** Employees vulnerable to phishing attacks.
- **No Intrusion Detection/Prevention System (IDS/IPS).**

Opportunities

- **Implement MFA:** Improve authentication security.
- **Security Awareness Training:** Reduce phishing risks.
- **Deploy SIEM System:** Centralized logging and monitoring.
- **Cloud Security Enhancements:** Use zero-trust architecture.
- **Penetration Testing & Audits:** Identify hidden vulnerabilities.

Threats

- **Phishing Attacks**
- **Ransomware**
- **Insider Threats**
- **Data Breaches**
- **Zero-Day Vulnerabilities**
- **DDoS Attacks**

2# Write a program to simulate simple brute force attack. Try to crack numeric passwords of length 4,6 and 8 digits.

```python
def brute_force(password):
    length = len(password)
    attempt = 0

    for num in range(10**length):
        attempt += 1
        guess = f"{num:0{length}d}"

        if guess == password:
            print("Password cracked!")
            print("Password:", guess)
            print("Attempts:", attempt)
            return

    print("Password not found.")

passwords = ["0001", "0123", "1234", "56289123"]

for pwd in passwords:
    print("\nTrying to crack:", pwd)
    brute_force(pwd)
```

```
Trying to crack: 0001
Password cracked!
Password: 0001
Attempts: 2

Trying to crack: 0123
Password cracked!
Password: 0123
Attempts: 124

Trying to crack: 1234
Password cracked!
Password: 1234
Attempts: 1235

Trying to crack: 56289123
Password cracked!
Password: 56289123
Attempts: 56289124
```

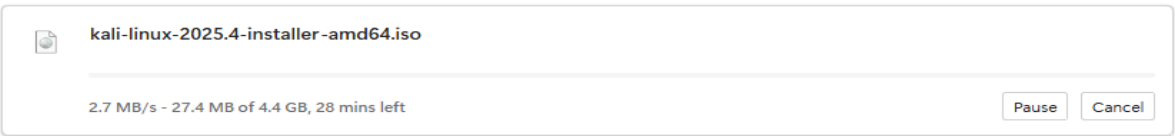
3# Install Kali Linux in your USB drive and demonstrate booting of the system using it. Write all the steps with proper screenshots.

Requirements:

- USB drive (minimum 8GB recommended)
- Kali Linux ISO file
- Rufus (Windows) or Balena Etcher
- A computer that supports USB boot

Step 1: Download Kali Linux

1. Go to https://www.kali.org
2. Download the latest Kali Linux ISO (64-bit Installer)



Step 2: Download Rufus (For Windows)

1. Go to https://rufus.ie
2. Download and open Rufus (no installation required)

Step 3: Create Bootable USB

1. Insert USB drive
2. Open Rufus
3. Select:
   - Device → Your USB
   - Boot selection → Select Kali ISO
   - Partition scheme → MBR (or GPT depending on system)
4. Click **Start**
5. Wait until process completes

Step 4: Boot from USB

1. Restart computer
2. Press Boot key (F12 / F9 / ESC / DEL depending on system)
3. Select USB device
4. Choose:
   - "Live system"
     OR
   - "Graphical Install"

Step 5: Install Kali (Optional)

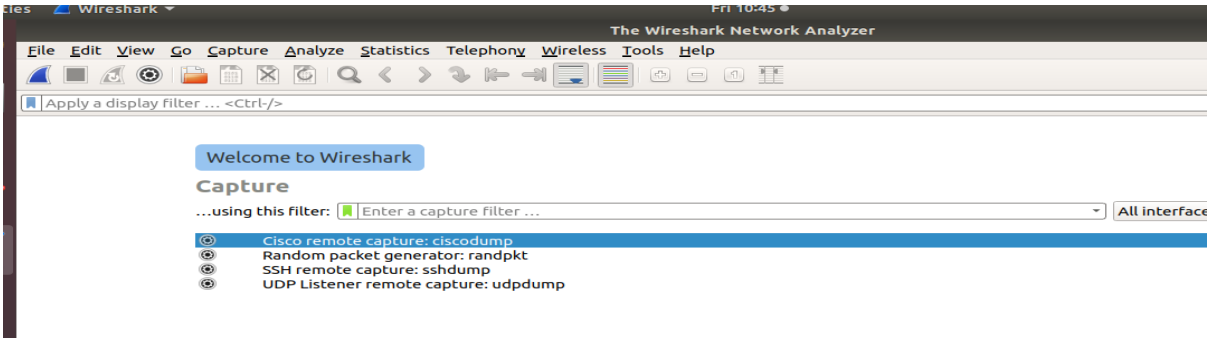
1. Choose Graphical Install
2. Select language
3. Configure network
4. Set username/password
5. Select disk (USB for persistence if desired)
6. Complete installation

4# Demonstrate the working of Wireshark tool to capture a sample TCP/IP packet details in Kali Linux. Write all steps with proper screenshots.

Step 1: Open Wireshark

1. Boot into Kali Linux
2. Open terminal
3. Type:
4. `wireshark`

   OR open from Applications → Sniffing & Spoofing → Wireshark



Step 2: Select Network Interface

1. Choose:
    - eth0 (wired)
    - wlan0 (wireless)
2. Double-click the active interface

Step 3: Generate Traffic

To capture packets:

- Open browser
- Visit a website (e.g., example.com)
- Or ping a server:
- `ping google.com`

Step 4: Filter TCP Packets

In filter bar, type:

```
tcp
```

Press Enter.

This shows only TCP packets.

Step 5: Inspect a Packet

1. Click on a TCP packet
2. Expand:
    - Ethernet II
    - Internet Protocol (IP)
    - Transmission Control Protocol (TCP)

You can view:

- Source IP
- Destination IP
- Source Port
- Destination Port
- Sequence number
- Flags (SYN, ACK, etc.)

Example TCP Handshake in Wireshark

You may observe:

1. SYN
2. SYN-ACK
3. ACK

This confirms TCP 3-way handshake.